

Recommendation stream by grain: OnlineRFPerm, RFPerm/CFPerm, and FSIDS

Y (outcome). One row is one order. $Y \in \{0, 1\}$ is **conversion** on that order (purchase / not). Y is the response only. It is never a column of X , never a ranking key, never a subset key.

X (covariates). X is the serving table of that order, sliced the way the log is written. Y is not in any slice.

grain	X columns	what the row is
order	amount, hour, n_items, channel	this order
merchant	merchant_cat, merchant_gmv, n_skus	the merchant on this order
user	user_tenure, user_hist_freq	the user on this order
all	the nine columns concatenated	anti-pattern; shown as a check

Region (south / north) is a merchant attribute, not a feature and not Y . The batch index T is a time label (D_{ref} vs the new batch), not a treatment.

Setup. $n_{\text{ref}}=400$, $n_{\text{new}}=120$, six batches, labeled onset $t=2$. After onset, only **south** merchants are shifted. Covariate / both plant amount $\succ$ merchant_gmv $\succ$ channel in $P(X)$. Concept plants amount in $P(Y | X)$ only. User columns never walk.

Table 1: OnlineRFPerm (Algorithm 1) — first rejection, delay, FAR.

Table 2: FSIDS / RFPerm ΔMSE / CFPerm φ -VIMP — Kendall- τ vs planted.

Table 3: south vs north, own-ref, last batch.

Table 1: WHEN. Frozen RF on D_{ref} . $T_t = \text{MSE}_t - E_{\text{ref}}$. Delay = first rejection — onset. FAR = mark before onset. Hop is a $1.5\times$ jump detector.

kind	grain	ADDIS	rank- p	SAFFRON	hop	last T	last MMD
covariate	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.067	0.095
covariate	merchant	5 ($d=3$)	2 ($d=0$)	5 ($d=3$)	miss	0.066	0.065
covariate	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	0.057	0.0006
covariate	all	FAR@0	FAR@0	FAR@0	miss	0.053	0.077
concept	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.113	−0.0005
concept	merchant	miss	2 ($d=0$)	miss	miss	0.019	−0.005
concept	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	0.009	0.0006
concept	all	FAR@0	FAR@0	FAR@0	miss	0.110	−0.001
both	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.157	0.095
both	merchant	miss	2 ($d=0$)	miss	miss	−0.019	0.065
both	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	−0.0005	0.0006
both	all	FAR@0	FAR@0	FAR@0	miss	0.170	0.077

Table 2: WHICH columns. Ranking recovery vs planted magnitude. User grain has no planted X columns — the correct negative control.

kind	grain	planted in X	FSDS recovered	τ_{FSDS}	τ_{RFPPerm}	τ_{CFPerm}
covariate	order	amount, channel	amount, channel	0.91	0.55	−0.18
covariate	merchant	merchant_gmv	merchant_gmv	0.82	−0.82	−0.82
covariate	user	—	—	—	—	—
covariate	all	amount, channel, gmv	amount, channel, gmv	0.69	0.18	−0.18
concept	order	amount	amount	0.24	0.24	0.71
concept	merchant	—	—	—	—	—
concept	user	—	—	—	—	—
concept	all	amount	amount	0.35	0.35	0.47
both	order	amount, channel	amount, channel	0.91	−0.18	0.91
both	merchant	merchant_gmv	merchant_gmv	0.82	0	−0.82
both	user	—	—	—	—	—
both	all	amount, channel, gmv	amount, channel, gmv	0.69	0.04	0.26

Table 3: WHICH accounts. Own-ref MMD: this region’s new bag vs this region’s D_{ref} . Subset key is region, not Y .

kind	grain	n_{south}	south MMD	north MMD	pair MMD	south $\Delta\mathbb{E}[Y]$
covariate	order	59	0.35	0.001	0.32	0.35
covariate	merchant	59	0.29	−0.005	0.29	0.35
covariate	user	59	−0.001	−0.001	−0.003	0.35
covariate	all	59	0.30	0.001	0.26	0.35
concept	order	59	−0.001	0.001	0.002	−0.041
concept	merchant	59	−0.010	−0.005	0.18	−0.041
concept	user	59	−0.001	−0.001	−0.003	−0.041
concept	all	59	−0.004	0.001	0.033	−0.041
both	order	59	0.35	0.001	0.32	−0.26
both	merchant	59	0.29	−0.005	0.29	−0.26
both	user	59	−0.001	−0.001	−0.003	−0.26
both	all	59	0.30	0.001	0.26	−0.26

Read-off. Order grain + ADDIS hits at the labeled onset ($d=0$). The concatenated **all** grain is FAR at $t=0$. User X does not move (MMD ≈ 0); a T mark there is Y walking through another grain. FSDS recovers the planted columns on the grain that actually contains them. South own-ref MMD is loud on covariate order (0.35 vs north 0.001); concept is quiet in X and shows up in $\Delta\mathbb{E}[Y]$.